

Measuring and Removing an Object’s Light Footprint: Reflection-Aware Object Removal in 3D Gaussian Splatting via Render–Edit–Refit

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Abstract

Deleting an object from a reconstructed 3D Gaussian Splatting (3DGS) scene removes its geometry but not its light footprint: the shadow it casts, its image in nearby mirrors, and the colour it bleeds onto surrounding surfaces all survive as painted-on texture. We study whether a pretrained video prior can remove this footprint through a simple *render–edit–refit* pipeline: render an orbit from the fitted scene, edit the video with a diffusion prior, and re-fit a fresh 3DGS on the edited frames. To measure the problem exactly, we built a synthetic benchmark in which the same camera path is rendered with and without the object, giving pixel-exact clean plates and automatically derived footprint regions. Our central negative finding is that a strong generalist editor (VACE) does not remove: it *substitutes*, replacing the masked object with a new one — and, given only the object mask, it reconstructs the object from its own shadow and reflection. A removal-specialised prior (ROSE) with masks covering the footprint behaves differently: footprint-region PSNR improves from 14.2 dB (deletion) to 28.4 dB. The masks need not be oracles — four clicks on a single frame, propagated by SAM 2, match oracle-mask performance (31.3 vs. 31.1 dB full-frame). Finally, the edits survive the lift back to 3D: all numbers are measured on held-out novel views rendered from the re-fitted scene, without any explicit multi-view consistency mechanism.

1 Introduction and Related Work

1.1 Problem

Imagine a red vase standing on a polished floor in front of a wall mirror. A 3D reconstruction of this room — here, a 3D Gaussian Splatting model [1] — stores the vase’s geometry, but it also stores everything the vase *does to the light*: the shadow it casts on the floor, its image in the mirror, the faint red glow it bleeds onto the wall. Formally, the radiance the scene records is

$$L = L_{\text{direct}} + L_{\text{obj}},$$

where L_{obj} collects every contribution that exists only because the object does. Removing the object’s Gaussians deletes the first-order geometry but leaves L_{obj} baked into the surrounding surfaces — a shadow with no caster, a reflection with no source. The goal of this project is to remove the object *together with* its light footprint, consistently across viewpoints, and to measure exactly how much of the footprint each method removes.

Our approach is deliberately simple: rather than reasoning about light transport inside the 3D representation, we render a smooth orbit from the fitted scene, hand the video to a pretrained

video diffusion prior that has seen millions of examples of scenes with and without objects, and re-fit a fresh 3DGS on the edited frames. The scientific question is whether such a prior actually knows how to *remove* — and whether per-frame edits survive the lift back to 3D.

Our contributions are:

1. **A controlled benchmark** with pixel-exact clean plates and automatically derived footprint regions, obtained by rendering the same camera path with and without the object (Section 2.2).
2. **A render–edit–refit pipeline** for footprint-aware object removal that requires no COLMAP, no model training, and no architecture changes (Section 2).
3. **An empirical study of video priors as removers**, with a clear negative result: a generalist editor substitutes rather than removes, to the point of rebuilding the object from its own footprint (Section 3.5).
4. **A practical annotation protocol**: four clicks on one frame, propagated by SAM 2, match oracle footprint localization (Section 3.7).

1.2 Related work

Object removal and 3D inpainting. Removal from radiance fields is usually cast as segmentation plus inpainting: SPIn-NeRF [2] lifts multiview masks and inpaints with a 2D prior; Weder et al. [3] learn view-consistent inpainting confidence; OR-NeRF [4] propagates user clicks with SAM [7]. In the 3DGS setting, GScream [5] and Gaussian Grouping [6] segment and delete Gaussians before repairing the hole. These methods focus on the hole the object leaves in geometry; the footprint on *other* surfaces is out of scope.

Footprint-aware removal. GauComp [8] and GOR-IS [9] explicitly target associated shadows during Gaussian removal, and full inverse rendering such as IRGS [15] would in principle explain reflections physically, at substantial cost in scene assumptions. Kuchler et al. [20] ask whether 2D editing models understand lighting at all. Our pipeline is representation-agnostic: all lighting reasoning is delegated to a video prior.

Measuring what removal leaves behind. SPIn-NeRF’s paired evaluation scenes and, most directly, Remove360 [13, 14] established that removal quality must be measured against ground truth *after* removal, and that residuals can remain semantically revealing. We follow this philosophy but move it to synthetic scenes where clean plates are pixel-exact and footprint regions are derived automatically rather than annotated.

Diffusion-based 3D editing. Instruct-NeRF2NeRF [10] and Instruct-GS2GS [11] alternate 2D edits (with InstructPix2Pix [16]) and 3D re-optimisation to keep multi-view consistency. Our pipeline is the degenerate one-round version: edit once, re-fit once; Section 3.6 measures whether that suffices.

Video diffusion priors. VACE [12] is a strong general video creation-and-editing model; ROSE [18] is fine-tuned specifically to remove objects *with their side effects* (shadows, reflections) within a single 2D video; $\int$ -noise [17] offers temporally consistent noise for video diffusion. ROSE removes side effects within a single 2D video; we study whether such a prior can remove them from a 3D *scene*, and measure what survives the lift back to 3D. SAM 2 [19] supplies the mask propagation for our practical protocol.

To our knowledge, no prior work provides a controlled benchmark that measures, per physical effect and in 3D, what object removal leaves behind — the gap this project addresses.

2 Method

2.1 Overview: render–edit–refit

Given a 3DGS scene containing an unwanted object, the pipeline is:

1. **Reconstruct** a standard 3DGS from the scene images (object present).
2. **Render an orbit:** a smooth camera path rendered from the fitted scene, so every frame comes with an exactly known pose.
3. **Edit the video** with a pretrained video diffusion prior under a binary mask (white = edit, black = keep).
4. **Re-fit** a fresh 3DGS on the edited frames with the unchanged poses.

We chose this design over the two main alternatives deliberately. Full inverse rendering [15] would model the footprint physically but constrains scene materials and lighting and is expensive to fit; iterative edit-and-optimize loops [10, 11] interleave many partial edits with 3D updates to enforce consistency, at several times the compute. Render–edit–refit is the cheapest member of this family — one edit, one re-fit — and part of our experimental question is precisely whether the expensive machinery is needed (Section 3.6). The pipeline trains no networks and needs no COLMAP: poses come from the orbit renderer.

2.2 Benchmark design

Measuring footprint removal requires knowing exactly what the scene looks like *without* the object — unknowable for a real scene, exact for a synthetic one. The benchmark is one compact Blender/Cycles scene — a red sphere on a floor, a white wall, and a wall mirror — rendered along an 81-frame, 120° camera arc at 832×480 (Cycles, 64 samples, denoised), twice: once with the object and once without, with identical cameras and lighting (Figure 1). The scene was iterated against an explicit checklist (every footprint effect must be visible to the eye), and several design choices carry the lessons of that iteration:

- **Polished dielectric floor, not a metallic mirror floor.** Our first floor was metallic: it produced a beautiful reflection and *no visible shadow*, because a pure specular surface has no diffuse response for a shadow to darken. A glossy dielectric floor (roughness 0.04) shows both a soft reflection and a clear shadow.

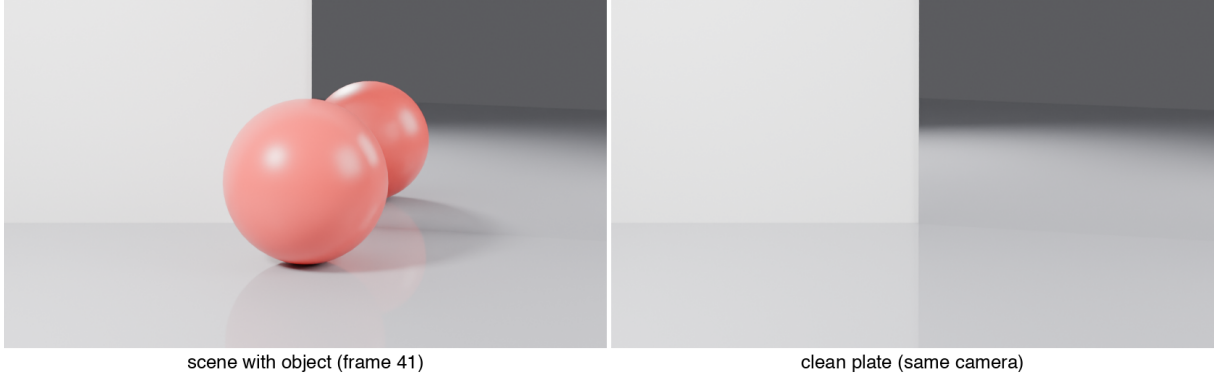


Figure 1: The benchmark pair at mid-arc. Left: scene with the object — note the four footprint components (shadow, mirror image, floor reflection, and a faint red bleed on the wall). Right: the exact clean plate from the same camera.

- **A separate vertical mirror panel** (metallic, roughness 0.01). At our viewing angles, dielectric Fresnel reflection on the floor is too weak to serve as the headline mirror effect; a dedicated mirror gives the footprint a second, unambiguous component.
- **A fill light.** A mirror faithfully shows the *unlit* side of the object; without a fill light the mirror image is a dark, unrecognisable blob. The key light is a small area light (harder shadow); the fill is weaker and sits on the mirror’s side.
- **Shared cameras $\Rightarrow$ exact clean plates.** Because the with/without renders share every camera, the without-render *is* the ground-truth answer to “what should removal produce?” Differencing the two renders outside the object silhouette yields footprint masks (shadow, mirror image, floor reflection, wall bleed) with no manual annotation.
- **832×480, 81 frames** matches the native resolution of the video models we evaluate and their $16n+1$ frame-count constraint, so the editor never resamples our data.
- **Every 8th frame held out.** All metrics are computed on the 11 held-out views, rendered from a re-fitted 3DGS — novel-view quality, not training-view quality. These views interpolate the arc; extrapolation far off the arc is not measured.

Object silhouettes are obtained by re-rendering the scene with only the object visible on a transparent background and thresholding the alpha channel at 0.5 — one Blender pass, no segmentation model in the loop.

2.3 Reconstruction and re-fitting

Both the initial fit and every re-fit use **splatfacto**, nerfstudio’s 3DGS implementation, with an identical recipe: 15,000 iterations, photometric loss

$$\mathcal{L} = (1 - \lambda) \mathcal{L}_1 + \lambda (1 - \text{SSIM}), \quad \lambda = 0.2,$$

following the original 3DGS formulation [1], with nerfstudio’s default densification schedule and learning rates. We use *random point initialisation* rather than an SfM point cloud: the poses are

exact and the scene is bounded, the regime in which the original 3DGS paper found random init competitive [1] — and it removes COLMAP from the pipeline entirely. The re-fit trains *from scratch* on the 70 edited training frames with the unchanged poses; no keep-loss or edit-region masking is applied, so any drift in unedited regions is deliberately left in and captured by the full-frame metric.

2.4 Video editors

We evaluate one generalist and one specialist editor, both small enough for a single Colab GPU:

VACE [12] (Wan2.1-VACE-1.3B): a DiT-based video diffusion model that conditions on a source video, a mask video, and a text prompt. We run 30 denoising steps at guidance scale 5.0 with a fixed seed, keeping the VAE at fp32 (bf16 VAE caused visible colour banding). Prompts describe the *target* state (“an empty room ... nothing on the floor”) and the negative prompt names the footprint effects (“red sphere, ... shadow of an object on the floor, pink glow”).

ROSE [18]: a removal-specialised editor built on Wan2.1-Fun-1.3B-InP (a video inpainting variant), whose transformer was fine-tuned by its authors on paired videos of scenes with and without objects, explicitly including side effects such as shadows and reflections. ROSE takes only video + mask (50 steps, the authors’ inference script and defaults); no prompt engineering is possible, which makes it a clean test of what the prior itself learned.

2.5 Mask protocols

The editors edit only inside the mask, so mask coverage is a first-class experimental variable. Three protocols, stated precisely:

- **Object mask:** the object silhouette (from the renderer’s index pass; in a real scene, SAM), dilated by 25 px. This is what a standard removal pipeline would use.
- **Oracle mask:** all pixels that differ between the with/without renders by more than 0.05 in any channel — object *and* footprint — union the silhouette, dilated by 15 px. This uses ground truth and is labelled an upper bound on localization.
- **Practical mask:** four clicks on one frame (object, mirror image, shadow, floor reflection), propagated bidirectionally across all 81 frames by the SAM 2.1 hiera-large video predictor [19], one tracked region per click; per-frame union of the four regions, dilated by 15 px.

2.6 Implementation

All code is in the submitted repository: a Blender/Cycles scene-and-dataset generator (`make_scene.py`, `extract_aux.py`) and five Colab notebooks covering reconstruction, deletion baseline, VACE editing + re-fitting + scoring, ROSE editing, and the SAM 2 protocol, plus a sixth notebook for the real-capture demo. The scene generator, the Gaussian-selection and deletion code, the mask construction, the scoring code, and all orchestration are written by the author; nerfstudio, diffusers/VACE, the ROSE inference script, and SAM 2 are used as unmodified third-party components. A `README` documents exact reproduction steps and the dependency pins that matter (a mid-project `transformers` upgrade silently broke ROSE’s loader — versions are now pinned in every notebook).

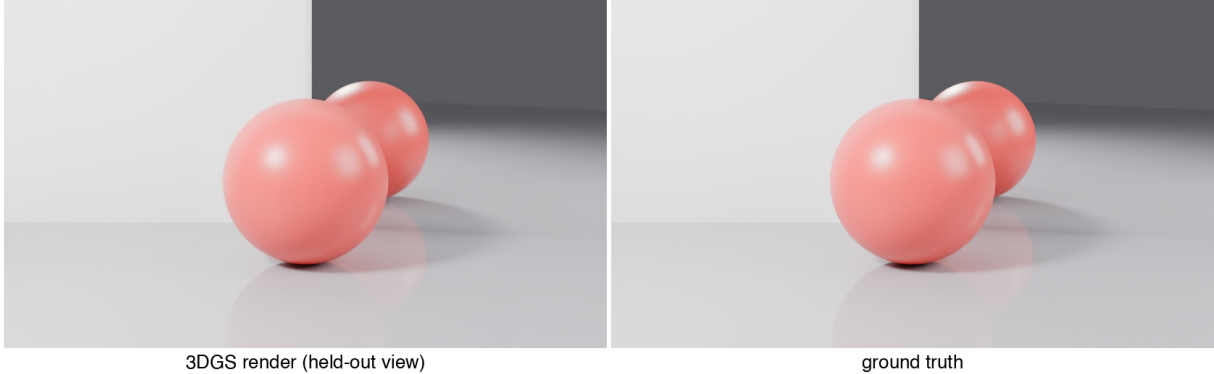


Figure 2: Reconstruction ceiling: a held-out view rendered from the fitted 3DGS (left) against ground truth (right) — visually indistinguishable at 45.7 dB.

3 Evaluation and Results

3.1 Metrics and protocol

Every experiment is scored the same way. The re-fitted (or edited) scene is rendered at the 11 held-out camera poses and compared against the exact clean plates:

$$\text{PSNR} = -10 \log_{10} \frac{1}{|\Omega|} \sum_{p \in \Omega} \|\hat{I}(p) - I_{\text{clean}}(p)\|^2,$$

with Ω either the full frame (*PSNR full*) or the ground-truth footprint region (*PSNR footprint*): pixels that changed between the with/without renders, excluding the object silhouette. Footprint PSNR is the honest number — the footprint occupies a minority of pixels, so full-frame PSNR can look respectable while the shadow sits there untouched. Scores are computed per held-out frame and averaged. All removal rows are measured *after* re-fitting a fresh 3DGS on the edited frames, i.e. they score what survives the lift to 3D, not the 2D edit. SSIM and LPIPS were computed for the reconstruction ceiling; for removal rows we report PSNR, which already separates the methods by an order of magnitude more than its noise.

3.2 Reconstruction quality (ceiling)

The Step-1 reconstruction reaches **45.7 dB** PSNR (0.992 SSIM, 0.021 LPIPS) on held-out views of the with-object scene (Figure 2). This matters for attribution: reconstruction error is negligible at the scale of every effect measured below, so any later degradation is caused by removal, not by 3DGS fitting.

3.3 The problem, quantified (M1: deletion)

The deletion baseline must first find the object’s Gaussians. We lift the 2D object masks: a Gaussian is selected if its center projects inside the mask in $\geq 90\%$ of the views that see it (≥ 40 views). Validated against the ground-truth sphere geometry, the 2,888 selected Gaussians achieve 100% precision and 90% recall; the missed 10% are 318 interior Gaussians invisible from outside, which a geometric check (known object placement, synthetic-only) adds to the

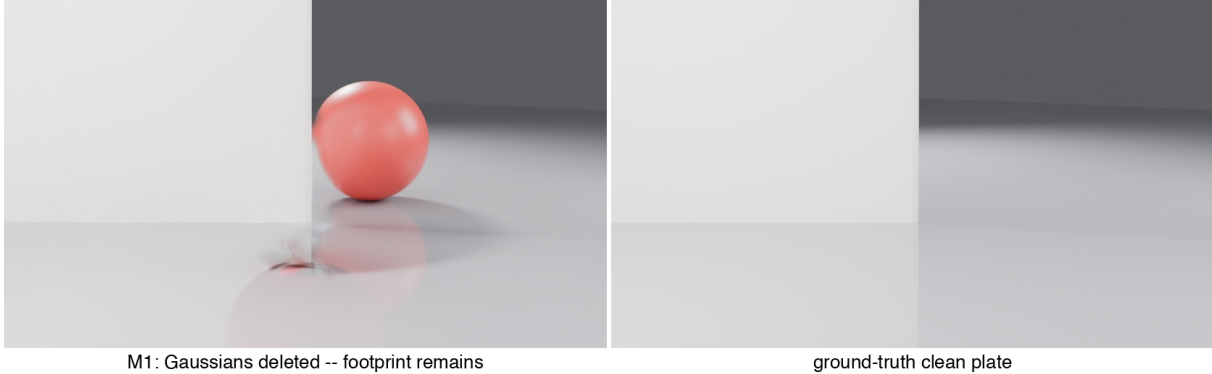


Figure 3: The ghost. Left: held-out view rendered after deleting the object’s Gaussians — a mirror image with no object, a shadow with no caster. Right: ground-truth clean plate.

deletion set. Deletion then does exactly what it promises: the sphere is gone, and its mirror image, shadow, and floor reflection remain, intact and sharp (Figure 3). Quantitatively: 23.1 dB full-frame, **14.2 dB** in the footprint — the number the rest of the paper tries to raise.

3.4 Main results

Method	PSNR full	PSNR footprint
Reconstruction ceiling	45.7	—
M1: Gaussian deletion	23.1	14.2
M3: VACE, object mask	21.6	14.2
M3: VACE, oracle mask	18.2	13.1
M3: ROSE, object mask	23.2	15.0
M3: ROSE, oracle mask	31.1	28.4
M3: ROSE, SAM2 masks (4 clicks)	31.3	28.4

Table 1: PSNR (dB) vs. exact clean plates on held-out novel views, full-frame and restricted to ground-truth footprint regions. All rows measured after re-fitting a fresh 3DGS on the edited frames.

Walking the table: the two VACE rows sit at or *below* deletion — the generalist prior actively hurts (Section 3.5 shows why). ROSE with the object mask barely moves the footprint number (15.0 dB): the specialised prior deletes cleanly inside its mask but touches nothing outside it, so masking — not the prior — is the bottleneck. Giving ROSE a mask that covers the footprint changes the regime: 28.4 dB in the footprint, a **+14.2 dB** gain over deletion, and 31.1 dB full-frame. The final row is the practical protocol: four clicks match the oracle (31.3/28.4), which is the operationally important result — oracle-quality localization costs a user seconds, not ground truth. As the benchmark is new, there are no published numbers to cite for it; deletion (M1) is the natural baseline all rows share.

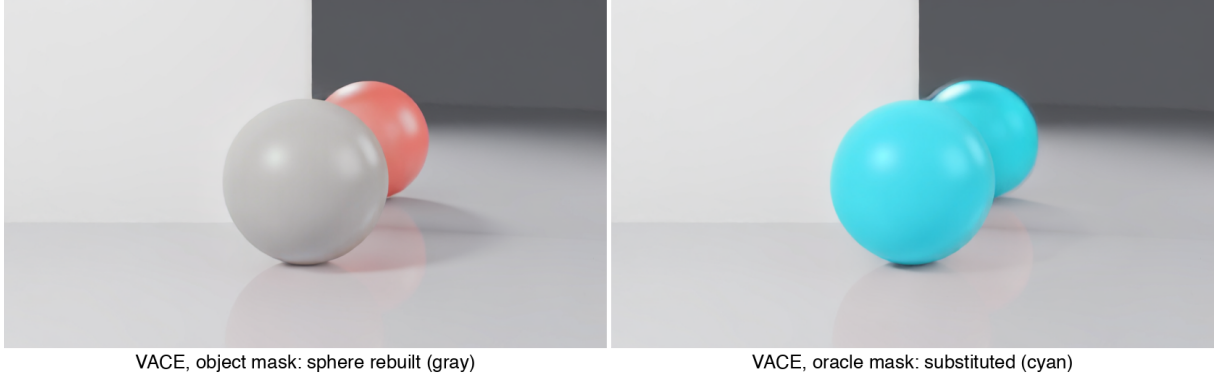


Figure 4: Substitution, not removal. Left: VACE with the object mask rebuilds the sphere (gray) from its own shadow and reflections. Right: with the oracle mask it substitutes a cyan sphere — with a physically coherent cyan mirror image. A third, emptiness-dominant prompt produced an orange sphere (not shown).

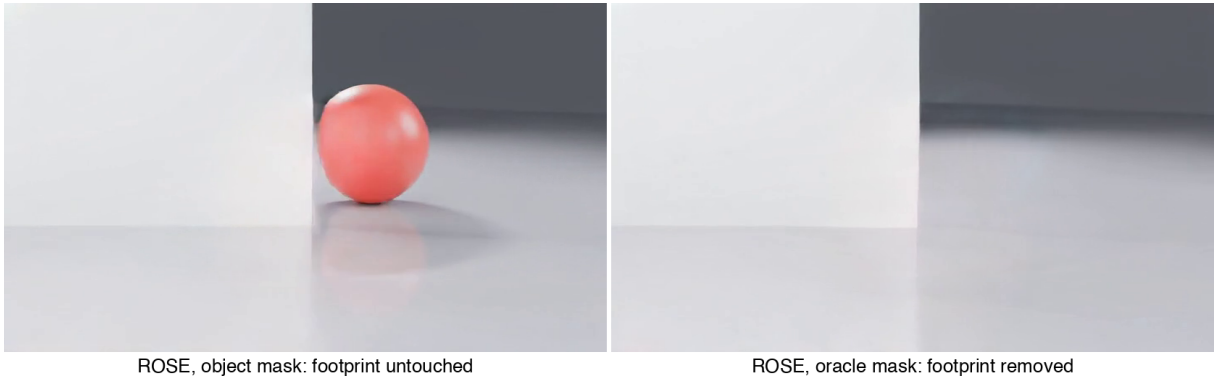
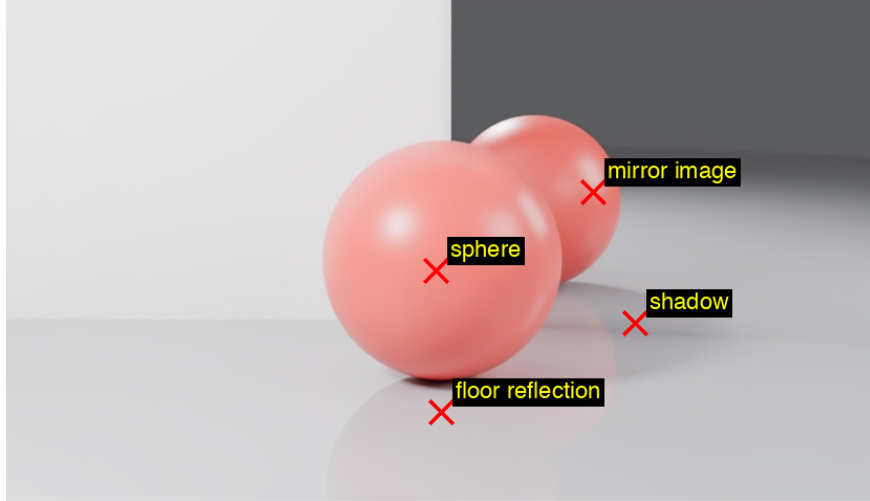


Figure 5: ROSE is mask-limited, not prior-limited. Left: object mask — the sphere is removed cleanly but its mirror image and shadow survive untouched outside the mask. Right: oracle mask — object and footprint removed together.

3.5 Why the generalist prior fails: substitution, not removal

VACE’s failure is systematic and instructive (Figure 4). Given the *object mask*, it filled the hole with a gray sphere — reconstructing the object it was asked to delete from the evidence left *outside* the mask: the shadow and reflections. This is direct visual evidence that the footprint encodes the object, the physical-scene counterpart of Remove360’s finding that residuals remain semantically revealing [14]. Given the *oracle mask* (footprint included), it substituted a cyan sphere — and rendered a *matching cyan reflection* in the mirror. The prior evidently understands object–reflection coupling; what it lacks is “generate emptiness” as an operation. Prompting cannot override the mask’s shape prior: across three prompt variants, including an emptiness-dominant prompt at raised guidance (50 steps, scale 7.5), we obtained three different objects (gray, cyan, orange) and zero removals. We conclude that the masked region’s shape acts as a strong object prior for a generation-trained model, and removal requires a prior trained for removal.



four clicks on the anchor frame (frame 41)

Figure 6: The entire annotation budget: four clicks on one frame.

3.6 Multi-view consistency of independent edits

The proposal reserved two remedies for cross-view inconsistency of per-frame edits: iterative dataset update in the style of Instruct-GS2GS [11], and warped noise [17]. The experiment resolved the question by measurement instead: ROSE edits all 81 frames of the orbit as a video but with no knowledge of our 3D scene, yet the re-fit on those frames reaches 31.3 dB on held-out views — above the edited-region quality that either remedy could plausibly rescue. At this operating point, the 3D representation itself absorbs the residual frame-to-frame disagreement, averaging it into a slightly soft but consistent scene. One honest caveat: at lower edit quality (e.g. VACE’s substitutions, which flicker in identity across frames) consistency machinery might still matter; we did not need it where it would have helped, and it could not have helped where we needed it.

3.7 The practical protocol

The oracle mask uses ground-truth differencing, unavailable in any real scene. The practical protocol replaces it with four clicks on the middle frame of the orbit — one each on the sphere, its mirror image, its shadow, and its floor reflection (Figure 6) — propagated bidirectionally across all 81 frames by SAM 2.1, unioned and dilated by 15 px (Figure 7). Two observations from running it: shadows and floor sheen are outside SAM 2’s comfort zone — the propagated regions drift over the arc — but the union over four regions plus dilation absorbs the drift; and the wall colour bleed was too faint to click at all, yet did not measurably hurt the score. The protocol ties the oracle (31.3 vs. 31.1 dB full-frame, 28.4 dB footprint in both cases), and the final result is an empty room whose mirror shows an empty room (Figure 8).

4 Limitations and Lessons

This study is deliberately narrow, and its limitations should be stated plainly. Everything quantitative is measured on a single synthetic scene with a single object; the camera arc spans

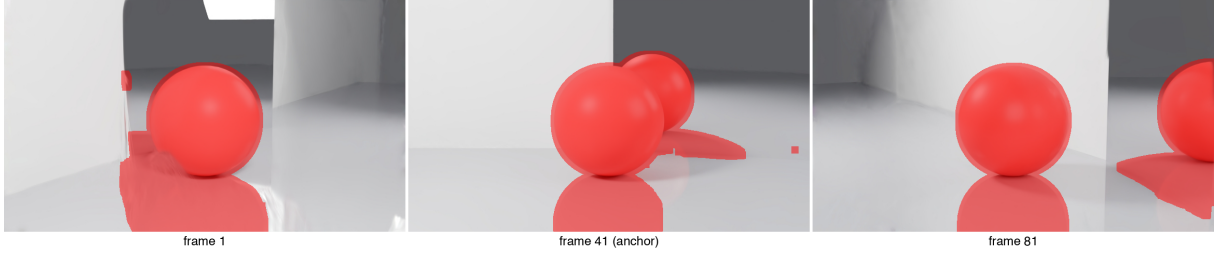


Figure 7: SAM 2 propagation of the four clicked regions (red overlay) at the start, anchor, and end of the arc.

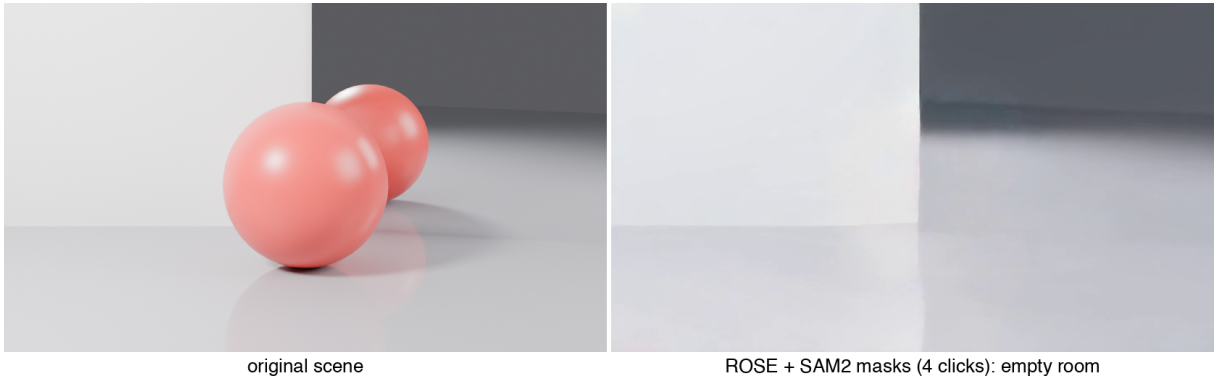


Figure 8: The headline result. Left: original scene. Right: ROSE with the 4-click SAM 2 masks — object, shadow, mirror image, and floor reflection all removed (31.3dB on held-out novel views after re-fitting).

120°, so held-out views interpolate rather than extrapolate. The re-fit uses no keep-loss, so unedited regions may drift — bounded, however, by the full-frame PSNR, which sits within 0.2dB of the footprint-driven gap. The oracle masks use ground truth, though the practical protocol matching them largely defuses this concern. Our metrics combine all footprint effects into one region; the per-effect masks (shadow vs. mirror image vs. bleed) already exist as a by-product of the differencing, and splitting the table by effect is mechanical future work, as is adding SSIM/LPIPS to the removal rows (currently reported for the ceiling only). The proposal’s in-context-examples extension (conditioning the editor on a Blender before/after pair) was not run: it was designed to teach a generalist prior what removal means, and adopting ROSE — a prior already trained for exactly that — made it moot; we report this as a descoped promise rather than a finding.

Two engineering lessons survived contact with the infrastructure: pin dependency versions inside every notebook (a transformers upgrade silently broke ROSE’s loader mid-project), and when an experiment looks absurd, verify *which inputs actually reached the model* before theorising — our first “VACE ignores the mask” hypothesis was in fact a mask-video encoding bug.

5 Conclusion

Removing an object from a 3DGS scene together with its light footprint can be done without touching the 3D representation’s internals: render an orbit, edit it with a removal-specialised video prior under masks that cover the footprint, and re-fit — raising footprint-region PSNR from 14.2dB to 28.4dB over deletion, with 31.3dB full-frame on held-out novel views and no explicit consistency machinery. The equally important negative result is that a strong generalist video editor cannot do this: it substitutes objects rather than removing them, and will happily rebuild an object from its own shadow. The benchmark — paired with/without renders, exact clean plates, derived footprint masks — is released with all scripts and notebooks at the project repository, and is, we believe, the right instrument for the next question: which physical effect (shadow, reflection, bleed) survives which remover.

Contribution Statement

All experiments and writing are by the author (solo project). Coding assistance and debugging support were provided by an AI assistant (Claude); all code was reviewed and all results verified by the author. Third-party models and libraries (nerfstudio, VACE, ROSE, SAM 2) are used as released, unmodified.

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